Mélissa COLIN

Deep Learning Architecture · Structured Attention · Constraints Built Into the Model

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Final-year MEng student in Computer Science (AI track, Engineer-Doctor track), graduating 09/27. I design architectures where the structure of the problem is written into the model rather than left to be learned from data: attention factorised along the axes that carry structure, and a differentiable constrained layer that makes the output valid by construction instead of merely penalised when wrong. Most recently at the University of Alberta on a Mitacs Globalink award.

Education

ENSEIRB-MATMECA, Bordeaux INP 09/24 – expected 09/27
MEng in Computer Science (diplôme d'ingénieur), AI track, Engineer-Doctor track (PhD preparation) *Bordeaux, France*

- Semester 8 (Feb–May 2026): 15.44/20, ranked 5th/81 (top 6%), up from 12.65 in semester 7. Semester 6: ranked 3rd/93.
- Deep Learning 18/20 · Research Initiation, PhD track 17/20 · Applied TCP/IP 17.5/20 · Cryptology 16.5/20 (French /20 scale; transcript attached).
- Final-year coursework (2026–27): large language models; knowledge and reasoning; data science and learning; cognitive architectures; AI applications (vision, music generation, AlphaGo).

EPSI Bordeaux 09/21 – 09/24
BSc in Artificial Intelligence and Data Science (RNCP level 6) *Bordeaux, France*

- Ranked 1st/10 in the final year (16.38/20).

Research Experience

Vision and Learning Lab, University of Alberta (Prof. Li Cheng) 05/26 – 09/26
Visiting Research Student, Mitacs Globalink Research Award *Edmonton, Canada*

- **Architecture.** Designed a permutation-equivariant transformer with attention factorised over three axes — people, joints, time — so that the symmetry of the problem is a property of the model rather than something the model has to infer from data. Physical interaction between people enters as an attention bias on the people axis.
- **Constraints inside the network.** Added a differentiable constrained layer as the final layer, trained end-to-end: the network proposes an output, the layer returns the nearest one that satisfies the constraints. Ground contact became a hard constraint instead of a penalty, leaving 0.001 mm residual violation on an input perturbed by 154 mm.
- **Making it trainable at scale.** Solved the inner constrained least-squares problem matrix-free (Gauss–Newton with preconditioned conjugate gradients), so the 47.9 GB dense Jacobian is never formed. Differentiated through the solver with the adjoint method rather than by unrolling it, and validated the gradients against finite differences to 6.6×10^{-10}. End-to-end inference 1.95× faster.
- **Architectural decision from a controlled experiment.** Measured that the first output parameterisation moved the result by only ∼2 mm, against 95% retained for the second. Traced the cause to a dimension mismatch — 6 equations constraining 159 unknowns — and redirected the architecture accordingly rather than tuning around the symptom.
- **Measurement discipline.** Found that run-to-run non-determinism from CUDA reduction order was 0.37 mm (median), larger than the effect I was trying to measure. Made training and evaluation deterministic at a 1.76× runtime cost, then re-ran the comparisons.
- **Data and compute.** Assembled a 26,000-clip corpus from several motion datasets, with pseudo-labels from three independent estimators; multi-day training over 14,912 windows on two RTX 4090 GPUs. An earlier refiner raised the success rate of the PHC physics tracker from 19.3% to 79.1% on 120 test windows (ground-truth motion scores 92.6%).

NAVER LABS Europe, co-supervised with Inria KINETIX 02/27 – 08/27 (accepted)
Research Internship — foundation models for 3D human perception (G. Rogez, J.-S. Franco) *Grenoble, France*

- Final research internship of the degree, on large pretrained visual encoders and what their representation can and cannot express about non-rigid scenes.

ENSEIRB-MATMECA, Research Initiation, PhD track (17/20) 02/26 – 05/26
Reproduction study of InterMask (ICLR 2025), a masked generative transformer *Bordeaux, France*

- Reproduced a discrete-token model — VQ-VAE tokeniser, then a transformer trained to fill masked tokens, the same modelling family as masked-token pretraining (FID 1.034 reported, 0.918 obtained).
- Tested two hypotheses of my own on top of it: an asymmetric contact penalty, and temporal data augmentation. Neither improved FID or R-Precision. I reported the negative result rather than selecting a favourable seed.

Publications & Awards

- **M. Colin** et al. A differentiable constrained layer for multi-person 3D motion refinement (University of Alberta placement). First-author manuscript in preparation. 2026
- Mitacs Globalink Research Award: international research funding for the University of Alberta placement. 2026
- **M. Colin**, I. Chraibi Kaadoud. *Performances and Explainability of ViT and CNN Architectures: An Empirical Study Using LIME, SHAP and Grad-CAM*. RJCIA @ PFIA (French national AI conference). First author, oral presentation. [HAL] 2024

Professional Experience

Sector Group 06/25 – 08/25
R&D Engineer, AI and Safety (internship) *Paris, France (remote)*

- Built a retrieval-augmented question-answering system over PDF documents with no text layer, for functional-safety work under IEC 61508.
- Worked on the decision to abstain, because a confident wrong answer cost more than a missing one: 100% precision on the answers returned, at 70% recall over 100+ evaluation queries. Audited 9 business units.

Cali Intelligences 01/23 – 09/24
AI Developer (apprenticeship); previously Applied AI Intern *Talence, France*

- Built the training pipeline for the real-time video detection models (YOLOv7, then YOLOv8): data ingestion, an augmentation library written from scratch in OpenCV, validation, YAML-driven orchestration. Raised detection accuracy from 60% to 90%.
- Corrected identity errors in multi-object tracking, halving false positives. Deployed with Docker and GitHub Actions; moved data storage to a bronze/silver architecture on AWS for GDPR traceability.

Skills

- **Programming languages:** Python (advanced), C, C++, SQL, Java, TypeScript/JavaScript, Bash.
- **Model architecture:** transformers and attention design, equivariance and weight sharing, VQ-VAE and discrete-token models, masked-token training, ViT and CNN backbones, LoRA/PEFT, retrieval-augmented generation.
- **Training and optimisation:** PyTorch, multi-GPU training, implicit differentiation through a solver, constrained optimisation, Gauss–Newton and conjugate gradients, deterministic evaluation.
- **Analysis and infrastructure:** ablation design, negative-result reporting, explainability (LIME, SHAP, Grad-CAM); Linux, Git, Docker, GitHub Actions, AWS.
- **Languages:** French (native), English (B2, TOEIC 840/990; four-month research placement conducted in English).

Leadership

- Vice-President, Forum INGENIB (Apr 2025 – present): EUR 100k budget, team of 20 students, 800 students and 80 companies hosted.
- Networking Lead, ai4industry workshop (Oct 2024 – Jun 2025): recruited and coordinated 22 speakers.